

DEPLOYMENT PLAN REPORT

Project Zync

The Sovereign AI for Rare Earth Engineering

UMHackathon 2026 Domain:

AI for Economic Empowerment & Decision Intelligence

Team: Gunners

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1. Executive Summary

This document outlines the industrial deployment strategy for Zync, a multi-agent AI infrastructure designed to unlock Malaysia's RM 1 trillion Rare Earth Element (REE) reserves. The architecture transitions from a prototype to a Sovereign Industrial Command Center, utilizing Z.ai GLM-5.1 for autonomous flowsheet reasoning while ensuring physical site safety through an integrated IIoT (Industrial Internet of Things) sensor layer. By bridging the Hydrometallurgical Proficiency Gap, Zync transforms raw geological dark data into ESG-compliant industrial blueprints.

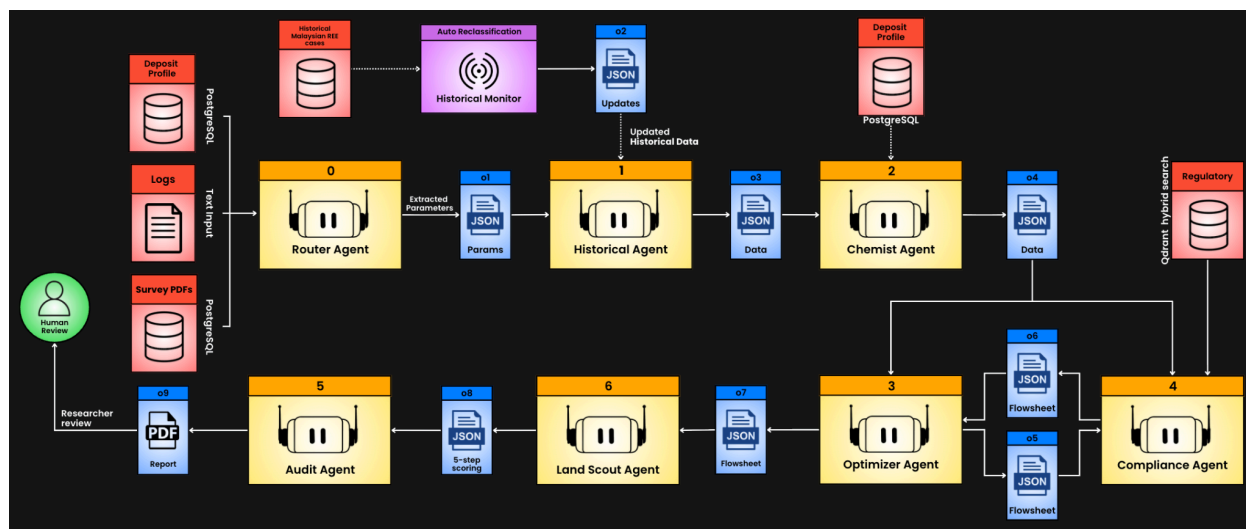
2. Infrastructure & System Topology

Zync adopts a sovereign-first topology to meet the strict data residency requirements of the **13th Malaysia Plan (13MP)**.

- **Deployment Model:** Utilization of On-Premise Local Server Infrastructure to eliminate foreign cloud dependencies and ensure 100% mineral data sovereignty.
- **Container Orchestration:** Utilization of Kubernetes (K8s) or Docker Swarm for automated scaling of independent microservices.
- **High Availability (HA):** Implementation of multi-node redundancy with a 99.9% uptime target, ensuring diagnostic records maintain operational continuity.
- **Security Layer:** Implementation of a Zero-Trust Security Model utilizing a local vault service for API key rotation and credential masking.

3. System Architecture

The Zync architecture is a specialized multi-agent ecosystem designed for high-stakes scientific reasoning and industrial optimization. It utilizes a Modular Multi-Agent Hub where task-specific agents coordinate to process both structured site data and unstructured geological reports.



3.1 The Agentic Pipeline

The system operates through a 7-stage agentic workflow:

- **Agent 0 (Router Agent):** Acts as the primary orchestrator, extracting parameters from Text Inputs, Logs, and Survey PDFs to delegate tasks to specialized agents.
- **Agent 1 (Historical Agent):** Interfaces with the Historical Monitor and Auto Reclassification modules to ground current site data in historical Malaysian REE cases.
- **Agent 2 (Chemist Agent):** Utilizes SciGLM to perform college-level scientific reasoning and numerical computing to predict reaction yields based on the Deposit Profile.
- **Agent 3 (Optimizer Agent):** Runs iterative loops to maximize mineral yield while minimizing cost and reagent consumption.
- **Agent 4 (Compliance Agent):** Performs Qdrant Hybrid Search against regulatory databases to ensure all proposed flowsheets meet ESG and AELB thresholds.
- **Agent 5 (Audit Agent):** Conducts a 5-step scoring process to verify the safety and integrity of the final industrial recommendation.
- **Agent 6 (Land Scout Agent):** Analyzes geological survey data to prioritize extraction zones based on strategic yield and regulatory risk.

3.2 Data Management & Retrieval

- **Structured Storage:** PostgreSQL manages persistence for site logs, deposit profiles, and operational audit trails.
- **Unstructured Retrieval:** Qdrant manages the vectorized "Dark Data" (Geological Surveys), enabling multi-hop retrieval through **GraphRAG**.
- **Researcher Review:** A "Human-in-the-Loop" gate allows for researcher review before the final PDF report is issued to the site command center.

4. Industrial IoT (IIoT) & Sensor Telemetry Layer

Zync bridges the cyber-physical gap by integrating real-time sensor data into the AI's causal reasoning engine.

- **Geotechnical Monitoring:** Deployment of sensors to monitor clay stability and prevent soil collapse during hydrometallurgical leaching.
- **Environmental Tracking:** Groundwater sensors monitor for leaching chemical (lixiviant) seepage to ensure ESG compliance.
- **AI Loop:** Telemetry is fed via MQTT protocol to the Agentic Hub, triggering an **Emergency Rollback (ER)** if structural pressure exceeds safety tolerances.

5. CI/CD/CT: High-Stakes Industrial Reliability

Zync adopts a Continuous Testing (CT) framework that treats AI reasoning as a critical industrial process. This ensures that the system's Sovereign Brain remains within safe hydrometallurgical and stoichiometry boundaries.

5.1 Scientific Reasoning Gates (CT)

Unlike standard software, Zync's automated pipeline includes a Scientific Reasoning Gate that audits every generated flowsheet before it reaches the production environment.

- **SFILES 2.0 Syntax Validation:** Every output from the Optimizer Agent (Agent 3) is parsed to ensure it adheres to the text-based chemical notation required for laboratory automation.
- **Stoichiometric Integrity Check:** The Chemist Agent (Agent 2) performs rigorous numerical computing to verify that mass and energy balances are preserved across the proposed process.

- **5-Step Scoring Audit:** The Audit Agent (Agent 5) executes a specialized scoring rubric—evaluating the flowsheet on chemical logic, economic feasibility, environmental impact, regulatory compliance, and structural risk.

5.2 Golden Release & Autonomous Interlock

To protect the **RM 1 trillion** mineral interest, Zync utilizes a Safety-First deployment protocol.

- **Golden Release Baseline:** A validated, rule-based Golden Version of the separation logic is held in a warm-standby state.
- **Autonomous Interlock:** If the Compliance Agent (Agent 4) detects a breach in AELB or DOE radioactive/chemical thresholds, the system triggers an immediate interlock.
- **Emergency Rollback (ER):** The interlock automatically reverts the site command center to the Golden Version, ensuring operational continuity while the anomalous AI reasoning is flagged for Researcher Review.

6. Sovereign Security & Governance

As Zync manages National Mineral Intelligence, the security framework is designed to prevent modern colonization through data leaks and to ensure 100% domestic residency.

6.1 Zero-Trust Security & Data Sovereignty

Zync operates on a Zero-Trust model, where every agent and sensor must be continuously verified.

- **Local Residency:** All inference from the Z.ai GLM-5.1 engine occurs within the On-Premise Local Server Infrastructure, ensuring no raw mineralogical data or survey PDFs leave Malaysian soil.
- **Credential Masking:** Utilization of a localized secure vault to manage Z.ai API keys and database credentials, preventing hardcoded exposure and unauthorized access to the reasoning engine.
- **Network Air-Gapping:** For high-stakes extraction zones, the system is configured to operate in an air-gapped environment with localized knowledge retrieval (GraphRAG).

6.2 Identity Governance & P1 Critical Control

Decision-making authority is strictly tiered to prevent industrial accidents and economic sabotage.

- **Role-Based Access Control (RBAC):** Defined roles, **Lau Hiap Meng (Team Lead)**, **Loh Yong Qiao (Backend/QA)**, and **Tai Jin Wei (Backend/QA)**, possess administrative oversight for system hardening.
- **P1 Batch Authorization:** Only certified **Process Managers** can authorize P1 Critical optimization batches that involve hazardous reagent concentrations.
- **Immutable Audit Trails:** Every action from **Agent 0's** parameter extraction to the **Land Scout Agent's (Agent 6)** zone prioritization is logged in a tamper-proof PostgreSQL audit table for national security review.

7. Operational Deployment Task List

The following schedule governs the site-level activation of the Zync platform for the industrial pilot phase.

Task	Start Date & Time	Finish Date & Time	Owner	Status	Comments
User Training (Site Engineers)	27/04/2026 09:00	28/04/2026 17:00	Penny Valencian	Completed	Training on Sovereign Command Center UI and feedback loops.
Stand up Production Environment	29/04/2026 08:00	29/04/2026 18:00	Loh Yong Qiao	Completed	Provisioning on-premise local server clusters.
Run Deployment	30/04/2026 08:00	30/04/2026 12:00	Tai Jin Wei	Completed	Automated deployment of multi-agent microservices.
Log Review & Reasoning Audit	30/04/2026 13:00	30/04/2026 18:00	Tai Jin Wei	Completed	Verifying Agentic Hub output accuracy against site baselines.

Final Production Safety Review	02/05/2026 09:00	02/05/2026 12:00	Lau Hiap Meng	Completed	Final audit of safety gates and regulatory thresholds.
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8. Strategic Conclusion

Zync is not merely a software tool; it is a **National Infrastructure Asset**. By leveraging **Agentic Engineering** within a safe, secure, and sovereign Malaysian framework, Zync is prepared to transform the REE sector from raw material export to high-value domestic production.